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Improving Two-Phase Flow Pattern Classification Accuracy in Horizontal Pipes Using Ensemble Techniques

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Abstract

This study explored the application of ensemble machine-learning models to predict two-phase flow patterns in horizontal pipes. Ensemble techniques including boosting, bagging, and random forests (RF) were employed. A novel decision-tree classifier was developed by combining Random Trees (RT), J48, Reduced-Error Pruning Decision Trees (REPT), Logistic Model Trees (LMT), and Naive Bayes (NBT) algorithms, specifically designed to predict flow regimes. The ensemble models were built using approximately 2380 experimental data points. Feature selection involved the application of six optimization methods, with training and cross-validation used for optimal selection. Performance evaluation utilized metrics like classification accuracy, precision, recall, confusion matrix, a measure of predictive performance, F-Measure, and Precision-Recall Curve (PRC) area. The study's results indicate that boosting and RF classifiers demonstrate superior prediction accuracy compared to other ensemble algorithms. Particularly, the RF model exhibited exceptional performance across various evaluation metrics, emphasizing the efficacy of ensemble algorithms over single classifiers for tree-based models in predicting flow regimes with an impressive accuracy rate of 91%. The findings underscore the consistent outperformance of ensemble models, which integrate multiple classifiers, over individual classifiers regarding prediction accuracy. This underscores the benefits of leveraging diverse models to enhance accuracy and robustness in flow pattern classification. The analysis also highlights the significant influence of volumetric liquid velocity and gas velocity as key features in determining the models' performance. This research presents a robust and refined ensemble approach, offering a cost-effective and highly accurate method for predicting two-phase flow regimes in horizontal conduits under various operating conditions.

Keywords: Two-phase flow, Horizontal pipes, Flow pattern classification, Ensemble Machine Learning

Introduction

Accurate prediction of two-phase flow patterns in horizontal pipes is vital for optimizing oil and gas transportation, power generation, and chemical processing. Traditional methods—visual observation, empirical correlations, and computational fluid dynamics (CFD)—face limitations in subjectivity, generalizability, and computational cost (Ruiz-Díaz et al., 2024). Recent advances in ML, particularly ensemble techniques, have shown promise in addressing these challenges.

A study by Alhashem, (2020) explored the ability to forecast multiphase flow patterns in horizontal pipes by supervised ML classification mechanisms with: liquid and gas viscosity, liquid and gas density, liquid and gas velocity, pipe diameter, absolute roughness, and surface tension. The data is gained from the Stanford Multiphase Flow Data (SMFD). In contrast, Support Vector Machine and Logistic Regression performed the best when there were numerous examples in a confirmed flow pattern and the worst when there were uncommon examples. Limited data availability made it unclear whether or not Neural Network MLP is affected. The best model performance for classifying flow regimes using the SMFD dataset was Random Forest, then Decision Tree.

This approach has been demonstrated to outperform traditional empirical methods and even standard machine learning models when applied to the prediction of two-phase flow rate through wellhead chokes in oil wells. The use of ensemble methods, along with deep learning algorithms, has been found to outperform traditional machine learning models in predicting two-phase flow patterns, showcasing the potential of these advanced techniques in improving predictive accuracy and performance (Barjouei et al., 2021).

Ensemble machine learning models have been used in several studies to predict two-phase flow patterns in horizontal pipes. (Ibrahim et al., 2022) developed a mathematical model using MATLAB to predict gas-liquid two-phase flow patterns in a horizontal pipe. (Wahid et al., 2023) investigated the performance of machine learning algorithms, including ensemble learning, to classify oil-water flow patterns in a horizontal pipe. (Soedarmo, 2023) compared the accuracy of a pure machine learning model, a hybrid model, and a high-definition physics-based model in predicting liquid holdup in gas-liquid stratified flow in pipes. (Cepeda-Vega et al., 2022) proposed a Generalized Additive Model (GAM) to predict pressure drop in gas-liquid two-phase flow in horizontal pipes. (Gong et al., 2023) explores the use of machine learning algorithms to improve the prediction of multi-phase pipeline transport. The developed models can be integrated with real-time data for better prediction and anomaly detection. The K-nearest-neighbors (KNN) model with $K = 50$ gave an optimal result with a 97.4% prediction accuracy for horizontal two-phase flow regime identification. The Multi-class support vector machine (MCSVM) model underperformed in terms of prediction accuracy (Manikonda et al., 2022).

(AL-Dogail et al., 2023) proposed an AI model using dimensionless parameters for accurate pressure drop estimation in two-phase horizontal flows. This method highlights the potential of AI for multi-phase flow prediction with minimal input requirements. Khan et al. (2023) provided a benchmark for future studies to use dynamic pressure for the prediction of a two-phase flow pattern. This research can be expanded by collecting data over a wider range of flow parameters and different geometries. The results display that the model used has high performance in two-phase flow regime prediction. The accuracies of identifications in stratified, slug, and annular flows are all above 80%.

(AL-Dogail and Gajbhiye, 2021) investigated the influence of fluid properties on two-phase flow regimes and pressure drop in horizontal pipes. Their work involved the development of a dataset encompassing the effects of density, viscosity, and surface tension. This dataset was generated through experiments conducted at varying liquid and gas velocities, alongside the aforementioned fluid properties. The generated data was then compared with existing models from the literature.

A recent study by (Hafsa et al., 2023) compared machine learning (ML) and deep learning (DL) algorithms, determining that the Extreme Gradient Boosting method (XGBoost) stands out as the best model for forecasting gas-liquid flow patterns. An effective two-phase flow regime prediction technique with high

performance is very critical since it will affect the thermal-hydraulic and heat transfer phenomena, and consequently, the safety of SMR. In (Huang et al., 2024) research, Random Forest (RF), K-Nearest Neighbor (KNN), and support vector machine (SVM) algorithm were selected as the candidates for ML techniques. Within the models they chose, RF and KNN have higher accuracies, both can reach more than 90 %. RF has the highest accuracy, which can reach more than 95 %. To further develop the performance of the prediction algorithm, the performance of both the RF and KNN is developed by the adoption of data preprocessing.

Study by (Arteaga-Arteaga et al., 2021) shows the performance of various artificial intelligence techniques to classify two-phase flow regimes, presenting the best alternates for this particular Classification problem using two-phase flow patterns (liquid and gas) in pipes. An incoming data analysis process using ML is suggested to quantify the forecast of gas-liquid two-phase regimes in pipes. This is achieved by contrasting the forecasting of obtainable methods versus an overall dataset collected from the literature, which includes 9,029 data points.

(Wahid et al., 2023) explored the efficacy of machine learning ensembles, specifically those incorporating weighted majority voting, in predicting flow patterns for oil-water mixtures within horizontal pipes. Their model achieved a high level of accuracy, exceeding 93%, by leveraging fluid and pipe properties as input data.

(Al-Sharifi et al., 2024) explored the potential of ensemble machine learning models for predicting two-phase flow regimes in vertical pipes. Their work reinforces the idea that tree-based ensembles offer superior robustness and accuracy compared to individual classifiers, aligning with the growing recognition of ensemble methods as a powerful tool for enhancing flow regime prediction.

Building on the work of (Khan et al., 2023), this study investigates the dynamic pressure behavior of two-phase flow in horizontal pipes. The goal is to identify key parameters that can be used to distinguish between different flow regimes. This approach leverages post-signal processing techniques, such as Fast Fourier Transform (FFT) and Probability Density Function (PDF) analysis, to extract informative features from the pressure data. This method offers a significant advantage for industrial applications as it eliminates the need for visual access to the flow, making it a practical and cost-effective solution.

This research paper breaks new ground by presenting a comprehensive evaluation of ensemble techniques for classifying two-phase flow patterns in horizontal pipes. The study identifies critical features for accurate prediction and delves into a detailed assessment of Random Forest's performance. Furthermore, a comparative analysis of boosting algorithms is conducted. These combined insights significantly advance our understanding of how machine learning can be effectively applied to multiphase flow systems.

Data Description

The current study utilizes a dataset of 2380 data points for training and testing ensemble models designed to predict flow patterns in a multi-phase flow system. The primary input features include liquid velocity, gas velocity, surface tension, fluid density, and fluid viscosity. The target variable includes 13 flow patterns, ranging from annular and slug to plug and dispersed bubbly flow. This variety of datasets allows the models to learn the complex relationships between flow parameters and pattern evolution. Figure 1 visualizes the distribution of data points across flow patterns.

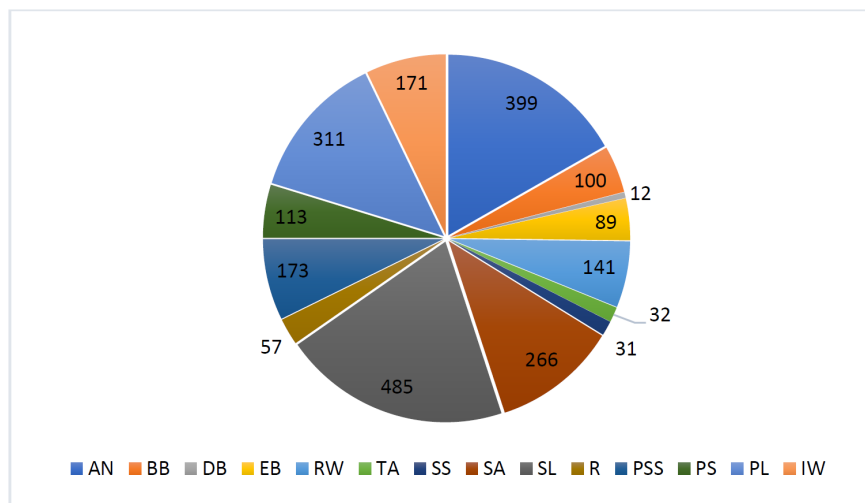


Figure 1—Distribution of data points by flow regimes

Methodology

Figure 2 outlines the key steps involved in building an ensemble model. This workflow diagram provides a structured approach for applying ensemble modeling to classify flow patterns in horizontal pipes. By leveraging the collective strengths of multiple models, ensemble methods offer a powerful tool for achieving reliable and robust flow pattern identification.

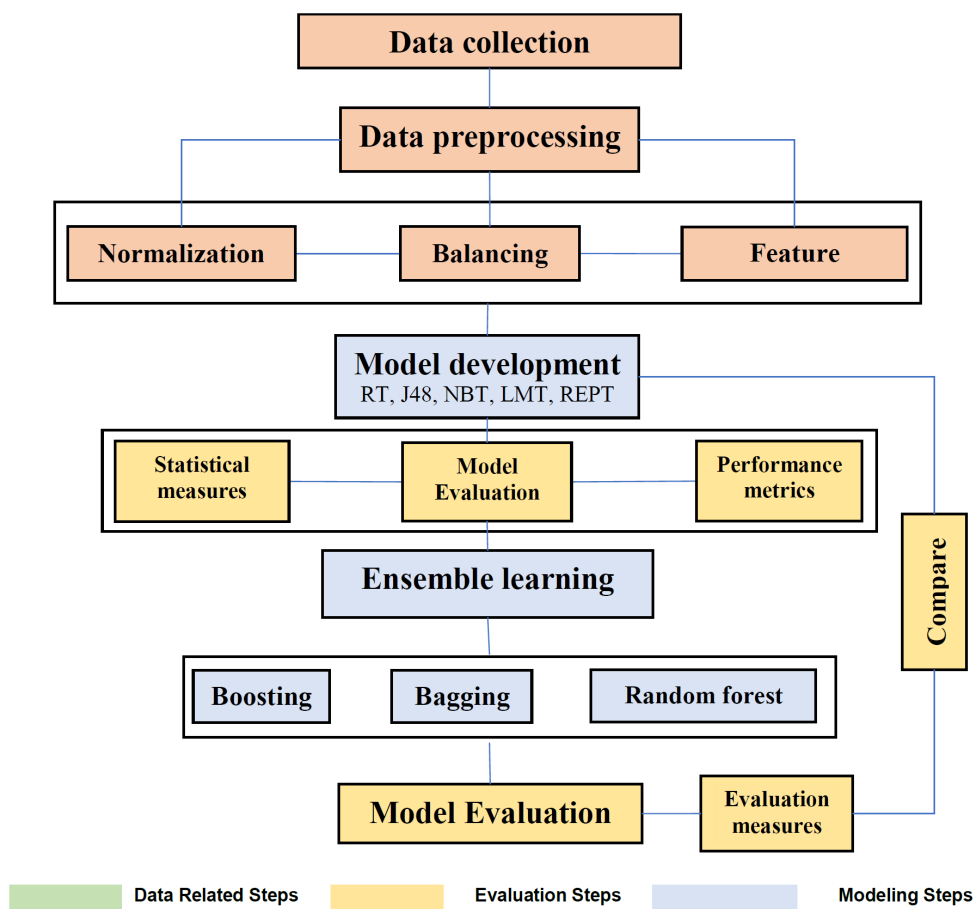


Figure 2—Workflow diagram for ensemble modeling process

Ensemble machine-learning models

Ensemble machine learning combines multiple base models to create a more robust and accurate predictive model (Lutins, 2017). The goal of using any machine learning technique is to find the one model that can predict the outcome the best. Rather than building one model and hoping it's the best/most accurate predictor, ensemble techniques take a wide range of models into account and average them to create one. Ensemble learning techniques can be divided into three categories: bagging, stacking, and boosting (Brownlee, 2021). Averaging, bagging, random forest, stacking, and boosting are some of the most used ensemble techniques (Mohammed and Kora, 2023). By developing advanced algorithms and generating results with a high degree of accuracy, ensemble approaches can aid in winning machine learning competitions. Voting and averaging are two of the easiest instances of ensemble learning in machine learning. Averaging is utilized for regression, while voting is used for classification. In several machine learning competitions, ensemble approaches have been successful and have been incorporated into the winning solutions (Demir, 2016). Powerful collaborative filtering algorithms have been implemented in the most well-known contests using ensemble approaches.

To improve the accuracy of models, ensemble approaches can lower bias and variance (Mohammed and Kora, 2023). Although there are essentially countless ways to combine models, three kinds of ensemble learning techniques are the most often explored and applied in real-world settings. Ensemble methods are a powerful technique that can improve the accuracy and performance of machine learning models. However, they come with a cost of increased complexity and expense, and the output of the ensembled model may be difficult to interpret and explain.

Results and Discussion

Feature Importance

When constructing a machine learning model, it is crucial to bear in mind that varying feature importance is merely one aspect to take into account. Figure 3 presents the relative contribution of each feature.

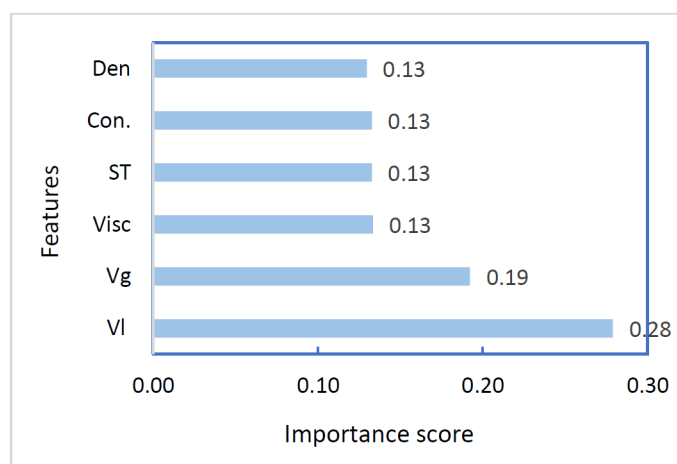


Figure 3—Feature importance of different parameters

The results presented in Figure 3 indicate that VI and Vg are the most influential features, with importance scores of approximately 0.28 and 0.19, respectively (Al-Dogail and Gaibhiye, 2021). Surfactant concentration ranks as the second most important feature, with an importance score of 0.28, followed by surface tension at 0.19. Liquid density has a moderate impact, scoring approximately 0.15, which is slightly higher than liquid viscosity (0.1), the least influential feature. Overall, surface tension and surfactant concentration demonstrate significant effects, while fluid density and viscosity exhibit minimal impact.

Impact of training parameters on model accuracy

Sensitivity analysis. In order to evaluate the individual and combined effects of different operational and physical parameters on flow regimes in horizontal pipes, the comprehensive experimental data set was strategically subdivided into distinct datasets as shown in Table 1. This data segregation facilitated a more focused analysis of the influence of each parameter on the observed flow behavior.

Table 1—Datasets with abbreviation and descriptions

Abbreviation	Descriptions
Set-1	Vg-VI
Set-2	Vg-VI-C-Den
Set-3	Vg-VI-C-ST
Set-4	Vg-VI-C
Set-5	Vg-VI-Den-Visc
Set-6	Vg-VI-Den
Set-7	Vg-VI-Visc
Set-8	Vg-VI-ST-Den
Set-9	Vg-VI-ST-visc

Key input parameters included: mixture type, superficial gas velocity (Vg), superficial liquid velocity (VI), temperature, pipe length, surfactant concentration, liquid density, and liquid viscosity.

Figure 4 illustrates how a Random Forest model's prediction accuracy varies depending on the dataset. Set-1 is the baseline dataset. The dataset Set-7 has the lowest accuracy (79.87%) of all the datasets, according to the plot. This implies that the viscosity data are critical to the model's accuracy. Additionally, the accuracy of the dataset Set-6 is slightly low (70.71%). This implies that the density data is crucial for the model's correctness as well. The accuracy of the Set-5 dataset yielded an accuracy of approximately 79.51%, suggesting that the included features moderately influence model performance. Out of all the datasets, Set-9 has the best accuracy (89.15%). This implies that even when the viscosity data is eliminated, the surface tension of the data may still contribute to increasing the model's accuracy. Another dataset with a high accuracy (87.78%) is Set-8. This implies that even after the density data is eliminated, surface tension may still contribute to increasing the model's accuracy. The accuracy of the Set-4 dataset is 88.62%. This implies that the viscosity and density data are more crucial for the model's correctness than the concentration data. The accuracy of the Set-3 dataset is 89.47%. This implies that even when the concentration data is eliminated, the surface tension of the data may still contribute to increasing the model's accuracy. The accuracy of the Set-2 dataset is 89.15%. This implies that when the concentration data is provided, the elimination of the density data has no appreciable effect on the model's accuracy. Overall, the sensitivity analysis demonstrates that the removal of the viscosity and density data has the greatest impact on the RF model's accuracy. Even with some of the data eliminated, the surface tension can aid in increasing the model's accuracy.

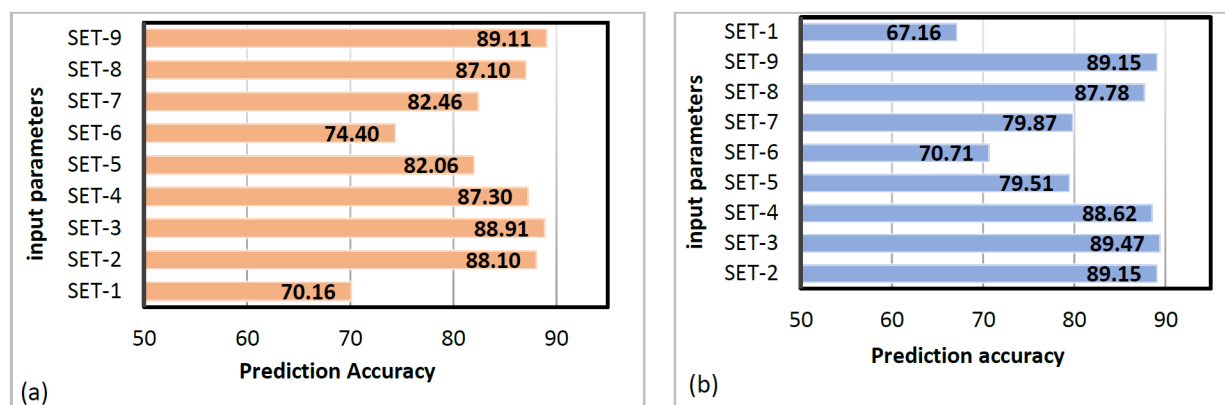


Figure 4—Sensitivity of prediction accuracy of RF model to different data set, {a) Training; b) Testing}

Evaluation of single machine learning model performance. Figure 5 shows the performance of several different machine learning models on a classification task. The models are J48, RT, NBT, REPT, and LMT.

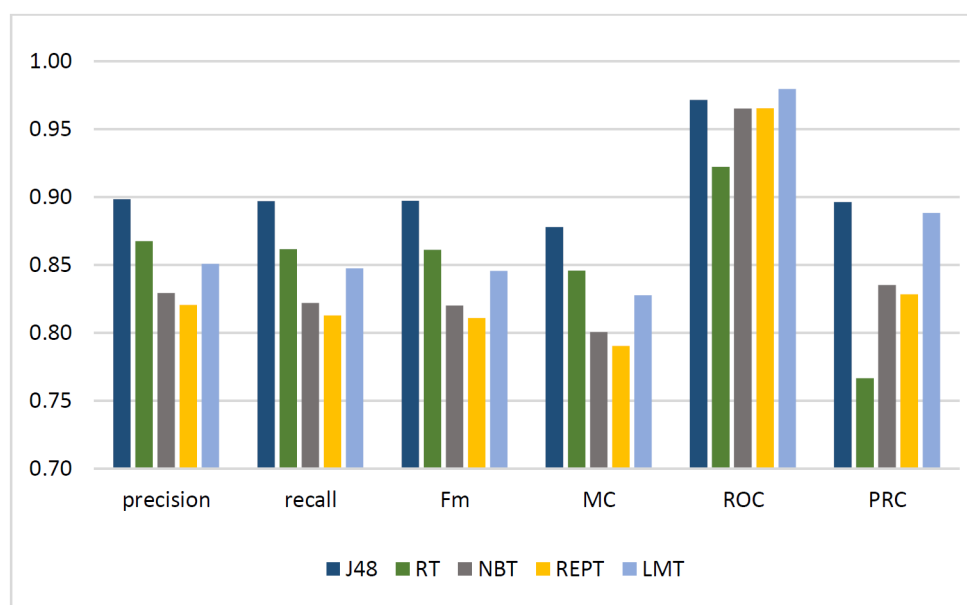


Figure 5—Evaluation measures for different models

This analysis examines the performance of various machine learning models based on the provided evaluation metrics. The key measures employed include precision, recall, F-measure (Fm), Receiver Operating Characteristic (ROC) Area Under the Curve (AUC), and Precision-Recall Curve (PRC) AUC. Among the evaluated models, the J48 decision tree algorithm emerges as the top contender, exhibiting superior performance across all three-core metrics (precision, recall, and F-measure). This indicates a well-balanced performance in correctly identifying positive instances while minimizing false positives and negatives. The high ROC AUC and PRC AUC scores of 0.97 further substantiate the J48 model's robustness in accurately predicting positive cases. While other models like Random Trees, Naive Bayes Tree (NBT), Reduced Error Pruning Tree, and Logistic Model Trees (LMT) demonstrate respectable performance, they fall short of the J48 model in terms of precision, recall, and F-measure. However, these models maintain competitiveness by achieving a reasonable balance between precision and recall, along with favorable Model Complexity (MC) and ROC AUC scores. This suggests their effectiveness in predicting positive instances.

In light of the comprehensive evaluation metrics, the J48 model stands out as the most reliable and robust performer. Its consistently high scores across various measures demonstrate its strong predictive power and well-balanced performance in classification tasks.

Performance evaluation of ensemble classification models. Al-Sharifi et al. (2024) demonstrated that tree-based ensembles excel in vertical pipe flow prediction, reinforcing their applicability in horizontal systems. Table 2 summarizes a comparative evaluation of several tree-based ensemble models employed for a classification task. The models' performance is assessed using a comprehensive battery of five metrics: precision, recall, F-measure, Matthew's correlation coefficient (MCC), and the area under the curve for both the ROC and Precision-Recall (PRC) curves.

Table 2—Summarizes a comparative evaluation of several tree-based ensemble models

		Precision	Recall	Fm	MCC	ROC	PRC
J48	Single	0.87	0.87	0.87	0.85	0.96	0.84
	Boosting	0.91	0.90	0.90	0.89	0.99	0.95
	Bagging	0.88	0.88	0.88	0.86	0.99	0.94
RT	Single	0.87	0.87	0.87	0.85	0.96	0.84
	Boosting	0.90	0.89	0.89	0.88	0.98	0.94
	Bagging	0.89	0.88	0.88	0.87	0.99	0.93
NBT	Single	0.83	0.82	0.82	0.80	0.97	0.84
	Boosting	0.91	0.91	0.90	0.89	0.99	0.94
	Bagging	0.86	0.86	0.85	0.84	0.99	0.93
REPT	Single	0.82	0.81	0.81	0.79	0.97	0.83
	Boosting	0.89	0.89	0.89	0.87	0.99	0.94
	Bagging	0.86	0.86	0.86	0.84	0.99	0.93
LMT	Single	0.85	0.85	0.85	0.83	0.98	0.89
	Boosting	0.90	0.89	0.89	0.88	0.99	0.94
	Bagging	0.88	0.88	0.88	0.86	0.99	0.94

The evaluation indicates that Model J48 achieved the highest performance among the single models evaluated. This was followed by LMT and RT, which demonstrated comparable performance. REPT and NBT, conversely, exhibited the lowest performance overall.

This research investigated the efficacy of ensemble learning methods compared to single tree-based models. The analysis revealed that ensemble techniques, including Boosting and Bagging, consistently outperformed individual models (J48, RT, NBT, REPT, LMT) across all evaluated metrics as shown in Figure 6. This finding suggests that ensemble methods have the potential to extract superior performance from the given dataset. Notably, Boosting emerged as the most effective ensemble approach, achieving the highest values for the majority of evaluation metrics across the tree-based models as shown in in Figure 6. While bagging also demonstrated strong performance, it generally fell short of Boosting's effectiveness.

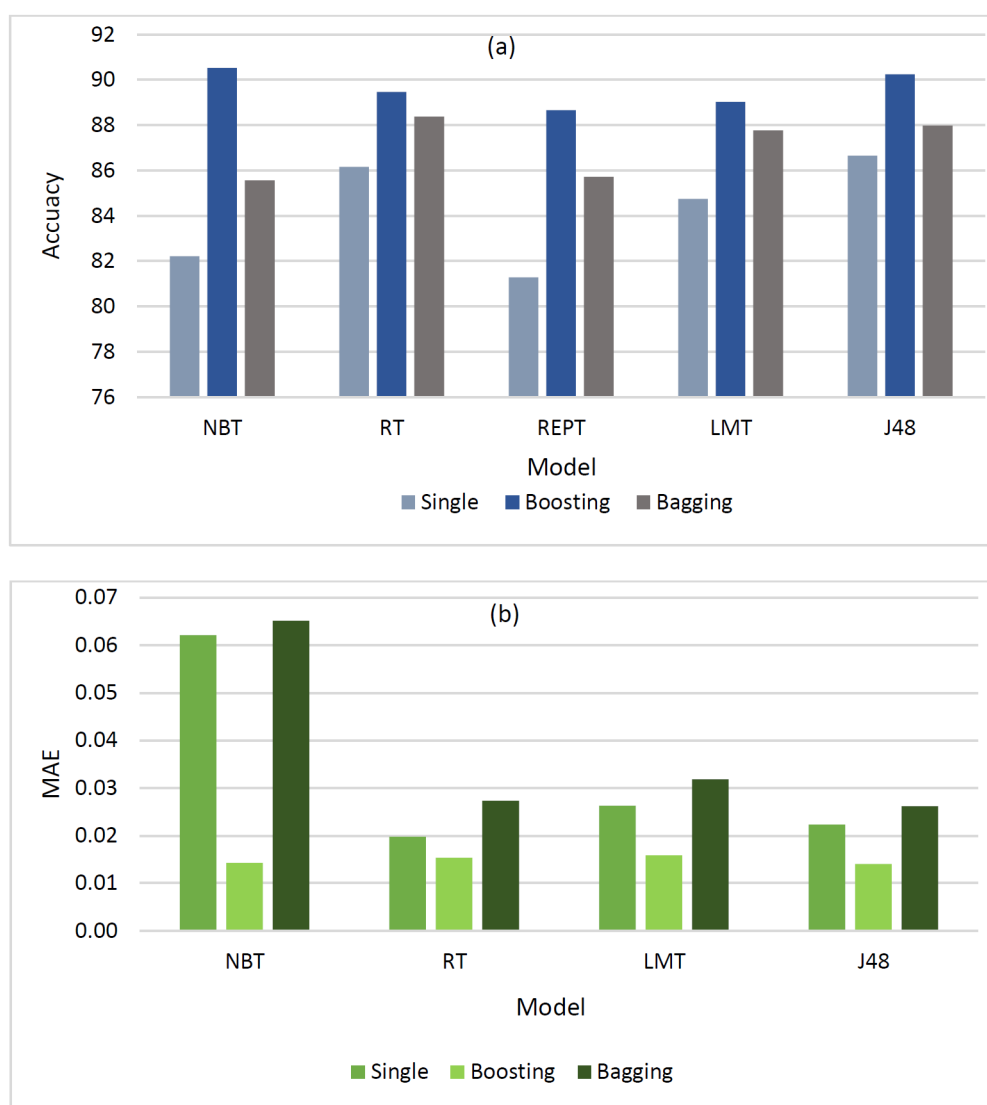


Figure 6—Comparison of classification performance metrics across different models. This figure depicts the accuracy (a) and Mean Absolute Error (MAE) (b) for various classification models.

The evaluation metrics demonstrate strong performance across all models in differentiating between positive and negative classes. This is evidenced by the high Area Under the ROC Curve and Area Under the Precision-Recall Curve scores achieved by all models. Furthermore, the Matthews Correlation Coefficient values for all models are also high, indicating that the models are not simply biased toward predicting the majority class as shown in Table 2. This study demonstrates that ensemble methods, specifically boosting, significantly improve classification performance. The models effectively differentiate between classes, exhibiting robustness against dependence on the majority class.

Boosting approach performance. Figure 7 presents the performance of different algorithms of boosting approach (Adaboost, Logitboost, and Multiboost). Their performance is measured by six metrics: accuracy, error rate, Kappa statistic, mean absolute error, root mean squared error, relative absolute error, and root relative squared error.

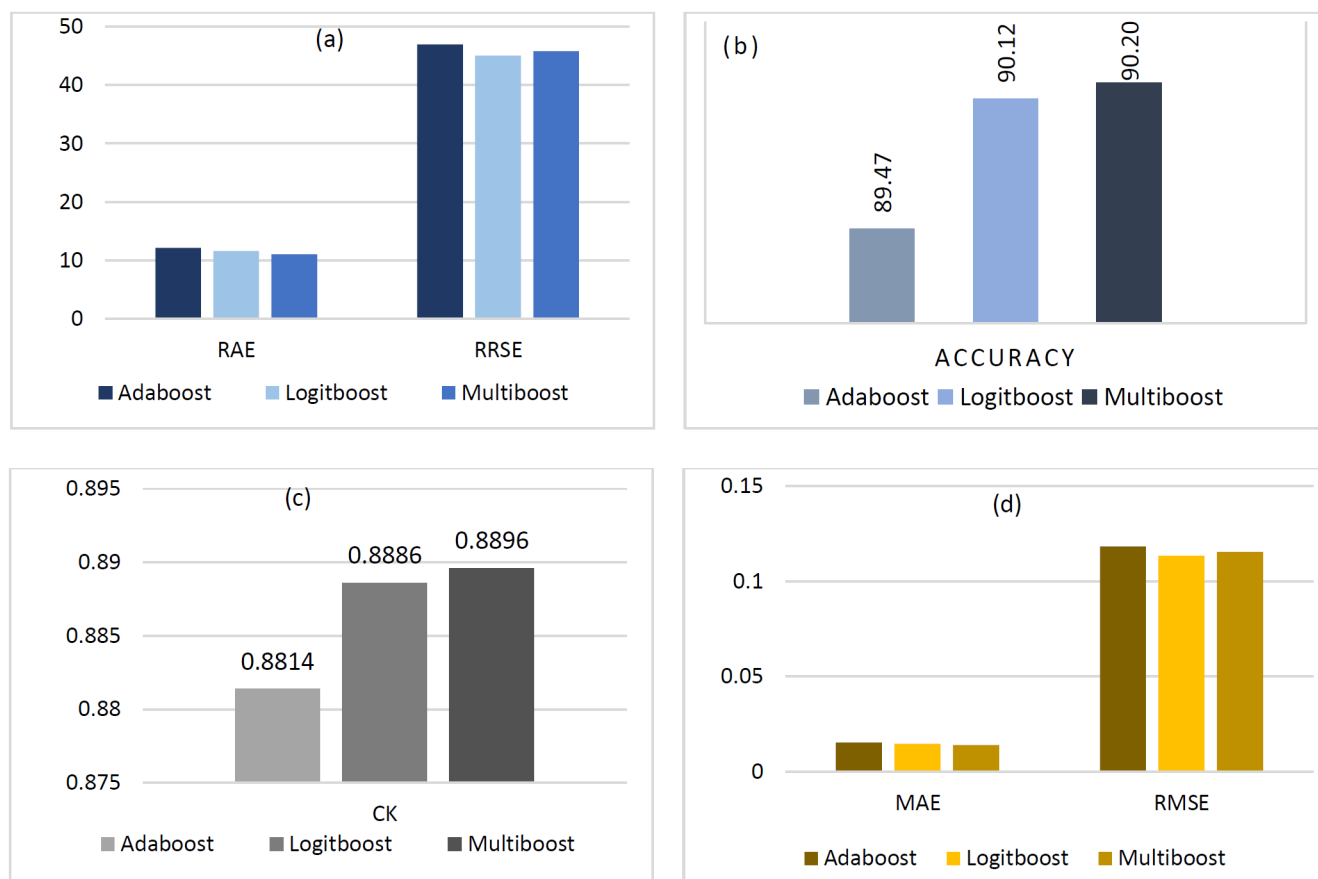


Figure 7—Evaluating algorithmic performance in Boosting Techniques. (a) RSE.RRSE; (b) accuracy; (c) CK; (d) MAE and RMSE

The comparative analysis of classification algorithms revealed Logitboost's exceptional performance across all evaluated metrics. Notably, it achieved the highest accuracy (90.12%) (Arteaga et al. (2021) followed by Multiboost (90.20%) and Adaboost (89.47%). This dominance extends to error rates, where Logitboost exhibited the lowest value (9.883) compared to its counterparts. Additionally, Logitboost demonstrated the strongest agreement beyond chance, as evidenced by its highest Kappa statistic (0.8886). Furthermore, Logitboost maintained its superiority in measures of prediction accuracy. It achieved the lowest Mean Absolute Error: 0.0147) and Root Mean Squared Error (RMSE: 0.1135), outperforming both Multiboost (MAE: 0.014, RMSE: 0.1155) and Adaboost (MAE: 0.0154, RMSE: 0.1184). Notably, Logitboost's dominance extended to relative error metrics, solidifying its position as the most accurate and precise model among those evaluated.

Performance comparison of single versus Bagged classification models. Figure 8 illustrates the impact of bagging on the performance metrics of various classification algorithms. Bagging, an ensemble technique, trains multiple iterations of a model on distinct data subsets. These models' predictions are then aggregated to enhance overall performance. Figure 8 reveals a consistent trend: all models exhibit higher accuracy compared to precision. Accuracy reflects the proportion of correctly classified true positives and negatives, while precision focuses on genuine identifications. A higher accuracy than precision suggests potential misclassification of negative instances as positive. Encouragingly, bagging generally improves model performance. This is evident in the increased precision, recall, F-measure, Matthews Correlation Coefficient, ROC Area ROC, and Precision-Recall Curve for all algorithms. Among individual models, J48 demonstrates the strongest performance across all metrics, with and without bagging. This suggests J48's inherent suitability for the given classification task. In conclusion, the findings support bagging as an

effective technique to enhance the performance of classification models. J48 emerges as the best-performing algorithm, further bolstered by Bagging's ensemble effect.

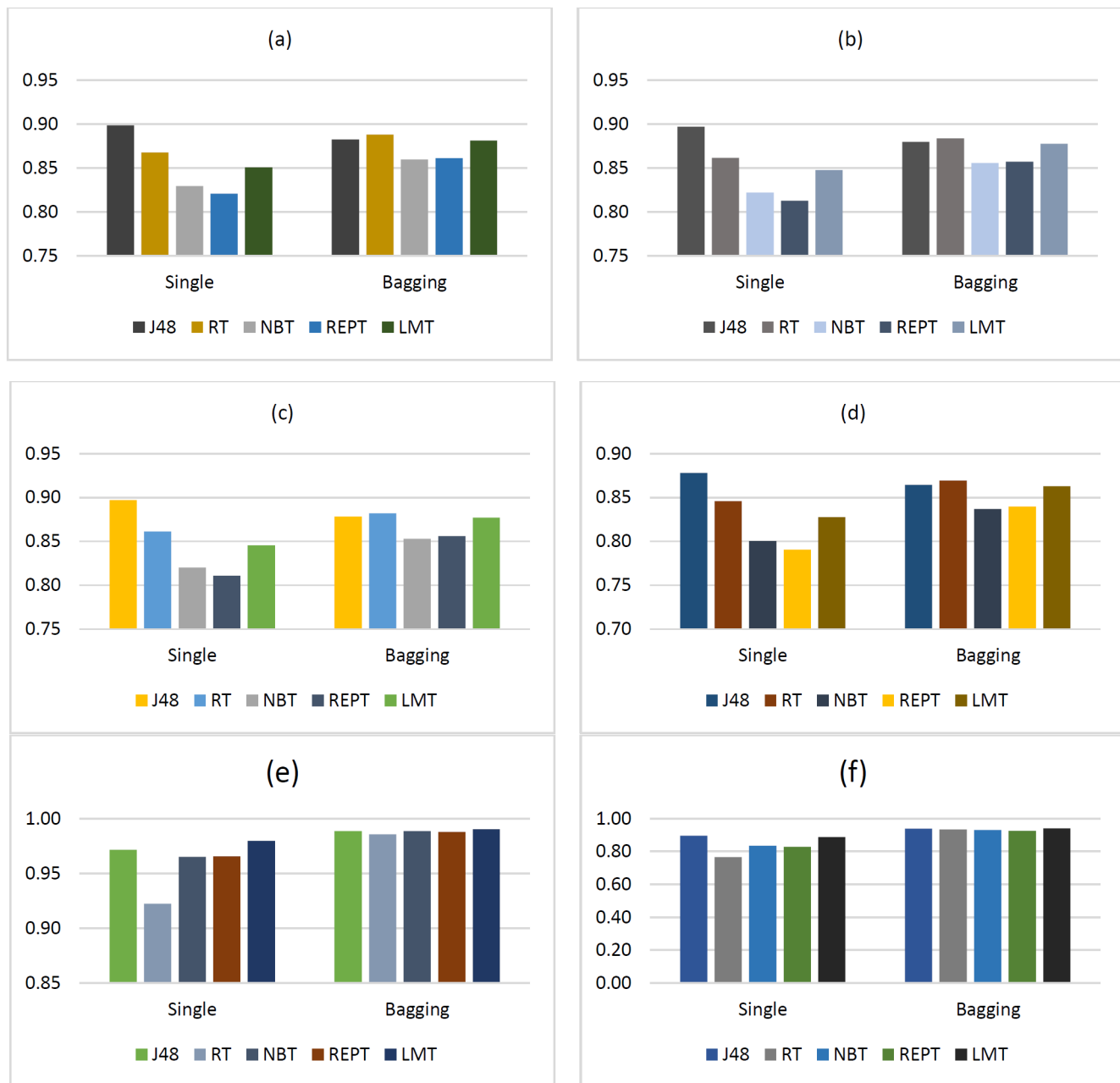


Figure 8—Impact of the Bagging approach on performance metrics across different machine learning algorithms, comparing them to single models. a) precision; b) Recall; c) F-measure; d) MCC; e) ROC; f) PRC

This study demonstrates that the Bagging ensemble method consistently outperforms single-model approaches across various performance metrics. Bagging models achieved statistically significant improvements in precision, recall, F-measure, Matthews Correlation Coefficient, Area Under the ROC Curve (AUC-ROC), and Precision-Recall Curve. Notably, the higher AUC-ROC and PRC values for Bagging models indicate a superior ability to balance true positive rates with false positive rates, leading to enhanced overall model performance and stability.

A comparison of the bagging ensemble learning approach and individual machine learning models is presented in Figure 9.

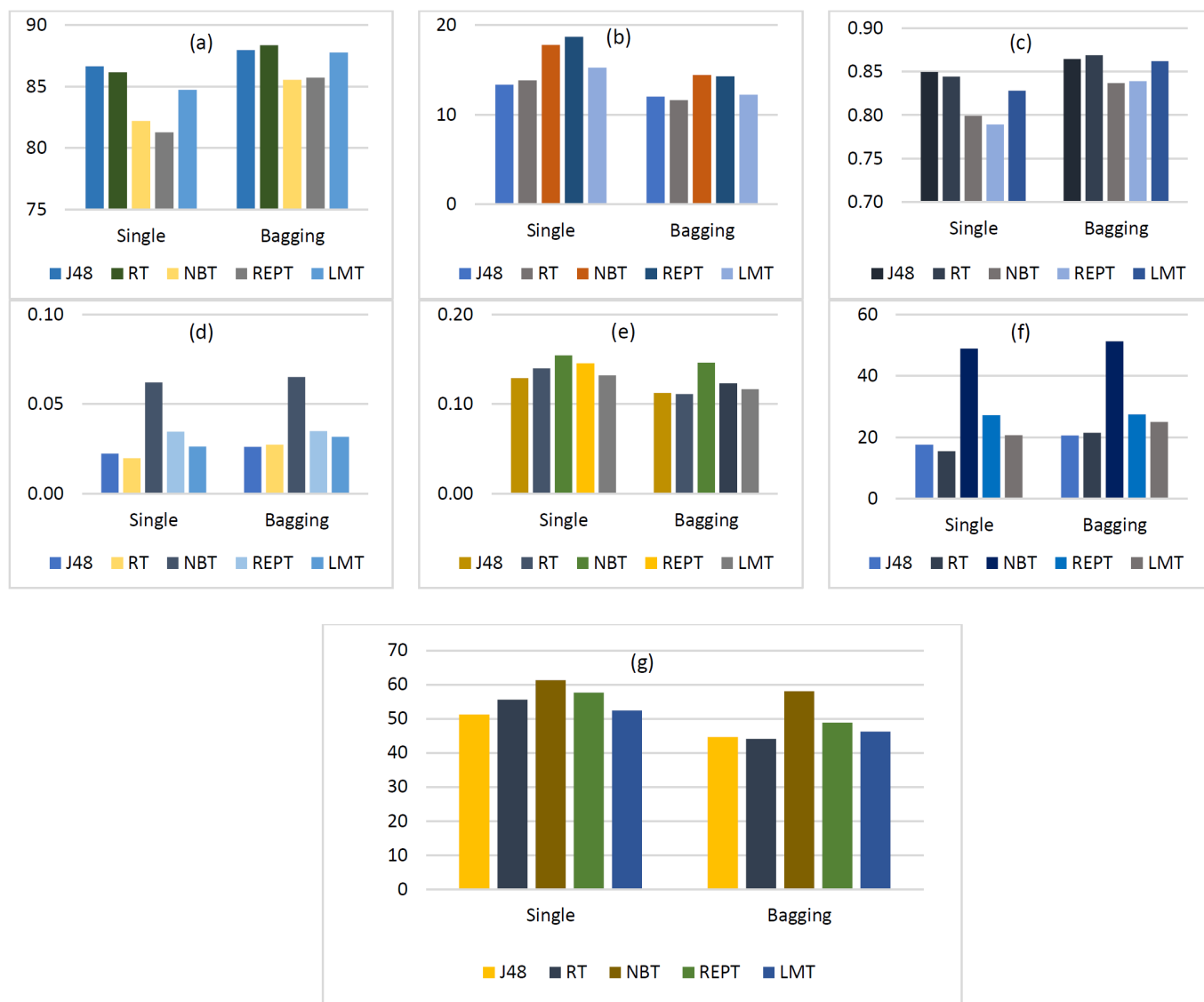


Figure 9—Performance metrics of single models versus models using the Bagging approach: (a) Accuracy, (b) Error, (c) KS, (d) MAE, (e) RMSE, (f) RAE, (g) RRSE

For each model type as shown in Figure 9 (J48, RT, NBT, REPT, LMT), the bagged ensemble version achieved a higher accuracy and lower error rate than the single model version. This suggests that bagging can be an effective technique for improving the performance of machine learning models. Both the single and bagged versions of these models achieved the highest accuracy (86.65% and 87.98% for J48, 84.75% and 87.78% for LMT) and lowest error rates (13.35% and 12.02% for J48, 15.25% and 12.22% for LMT). This suggests that these models may be good choices for the task at hand. The difference in performance between single models and bagged ensembles is small for some models. For example, the accuracy improvement for bagging J48 is only 1.33% (86.65% to 87.98%) and the error rate reduction is 1.33% (13.35% to 12.02%). This suggests that bagging may not provide a significant benefit for all models.

The specific performance metrics used in Figure 9 can provide different insights. For example, accuracy focuses on the number of correct predictions, while error rate focuses on the number of incorrect predictions. MAE and Root Mean Squared Error (RMSE) provide measures of the magnitude of the errors. RAE (Relative Absolute Error) and RRSE (Root Relative Squared Error) are normalized error measures that can be useful for comparing models on different scales.

Performance evaluation of the random forest ensembles model. Figure 10 shows the performance metrics precision, recall, and F-measure for a Random Forest model on training and cross-validation datasets. Precision is the proportion of true positives among the predicted positives, while recall is the proportion of true positives among the actual positives. The F-measure is a harmonic meaning of precision and recall and is used to balance the two metrics.



Figure 10—Performance evaluation of the random forest classifier: Fm, Precision, and Recall

As shown in Figure 10 the RF model has very high precision and recall on both datasets, with values of 0.999 for TR and 0.91 for CV. This means that the model is very good at correctly identifying both positive and negative instances. The F-measure is also very high for both datasets, at 0.999 for TR and 0.91 for CV. This further confirms that the model performs well on both datasets. It is concluded that based on the evaluation measures, the RF model appears to be a good choice for the task. Table 3 compares the performance metrics obtained during RF model training with those achieved through 10-fold cross-validation.

Table 3—Evaluation of RF model performance with (TR&CV)

Measure	TR	CV
Accuracy	99.88	91.12
Error	0.121	8.875
KS	0.999	0.899
MAE	0.011	0.027
RMSE	0.043	0.102
RAE	8.59	21.44
RRSE	17.00	40.61

The analysis as shown in Table 3 reveals a significant discrepancy between the performance of the Random Forest model on the training data (TR) and the cross-validation data (CV). The model achieves an accuracy of 99.88% on the training data, but this success diminishes considerably on the cross-validation data, where accuracy falls to 91.12% (Huang et al. (2024)). This substantial drop in accuracy is a strong indicator of overfitting (Brownlee (2021)). In simpler terms, the model appears to be memorizing the specific patterns within the training data rather than learning generalizable rules that can be applied effectively to unseen data. This is further supported by the error rates, with a training error of only 0.12% compared to a considerably higher error of 8.88% on the cross-validation data.

The model has a KS of 0.998 on the training data and 0.8986 on the cross-validation data. The decrease in KS between the training and cross-validation data suggests overfitting. The model has an MAE of 0.0109

on the training data and 0.0272 on the cross-validation data. The increase in MAE between the training and cross-validation data suggests overfitting. The model has an RMSE of 0.0428 on the training data and 0.1023 on the cross-validation data. The increase in RMSE between the training and cross-validation data suggests overfitting. The model has an RAE of 8.58 on the training data and 21.4408 on the cross-validation data. The significant increase in RAE between the training and cross-validation data suggests overfitting. The model has an RRSE of 17 on the training data and 40.6129 on the cross-validation data.

Analysis of the confusion matrix. The confusion matrix for the RF ensemble model is presented in Table 4. The diagonal elements of the confusion matrix, highlighted in Table 4, indicate the number of correctly classified instances for each flow type. These values reveal that the model performs best at predicting "Annular" and "Bubble" flows. Conversely, the model encounters greater difficulty in classifying "Slug" and "Annular Slug" flows (Khan et al. (2023)). This is evident from the higher number of misclassifications for these types, as they are frequently confused with "Plug" and "Pseudo Slug" flows, respectively. Additionally, the model exhibits challenges in differentiating between "Dispersed Bubble" and "Elongated Bubble" flows, as reflected by a significant number of inter-class misclassifications.

Table 4—Confusion matrix for flow type prediction

	AN	BB	DB	EB	RW	TA	SS	SA	SL	R	PSS	PS	PL	IW	
AN	373	0	0	0	0	0	0	0	0	0	0	21	0	5	AN
BB	0	97	2	0	0	0	0	0	0	0	0	0	0	1	BB
DB	0	2	46	0	0	0	0	0	0	0	0	0	0	0	DB
EB	0	0	0	84	0	3	0	0	0	0	0	0	2	0	EB
RW	0	0	0	0	163	0	0	0	2	4	0	0	2	0	RW
TA	0	2	1	0	1	286	3	12	0	0	6	0	0	0	TA
SS	0	0	0	0	0	9	97	4	0	0	3	0	0	0	SS
SA	0	0	0	0	0	10	0	139	0	0	21	0	2	1	SA
SL	0	0	0	0	0	0	0	0	55	2	0	0	0	0	SL
R	0	0	0	0	6	1	0	0	4	130	0	0	0	0	R
PSS	0	2	0	0	0	2	1	7	0	0	458	15	0	0	PSS
PS	30	0	0	0	0	0	0	0	0	0	21	215	0	0	PS
PL	0	0	0	0	5	0	0	0	0	0	0	0	57	0	PL
IW	5	3	0	0	0	0	0	0	0	0	0	0	0	56	IW

The confusion matrix unveils a critical limitation of the model: its proneness to misclassifications despite achieving some success in flow type prediction. Delving deeper into the features associated with these errors is crucial for enhancing model performance.

ROC curve. The area under the receiver operating characteristic curve is a widely used metric to assess the performance of binary classification models. It reflects the model's ability to discriminate between positive and negative classes across various classification thresholds. A higher AUC value generally indicates better classification performance, but it's important to consider other evaluation metrics alongside AUC for a comprehensive assessment. Figure 11 illustrates the ROC curves generated by the RF model for each identified flow pattern. These curves allow us to evaluate the model's performance in differentiating between normal and anomalous flow patterns for each specific pattern type.

Visual inspection of the ROC curve (Figure 11) reveals an ideal performance profile, where the curve closely follows the upper left corner of the graph. This indicates a high True Positive Rate (TPR), signifying the model's ability to correctly identify positive cases, coupled with a low False Positive Rate (FPR), reflecting minimal misclassification of negative cases. The Area Under the Curve of the ROC chart, at approximately 97%, further underscores the model's robustness in distinguishing between positive and negative classes. Similarly, the ROC curve for Random Forest demonstrates excellent performance for

classifying BB, EB, and DBB, with the curve situated in the upper left corner and achieving a high AUC of 0.998.

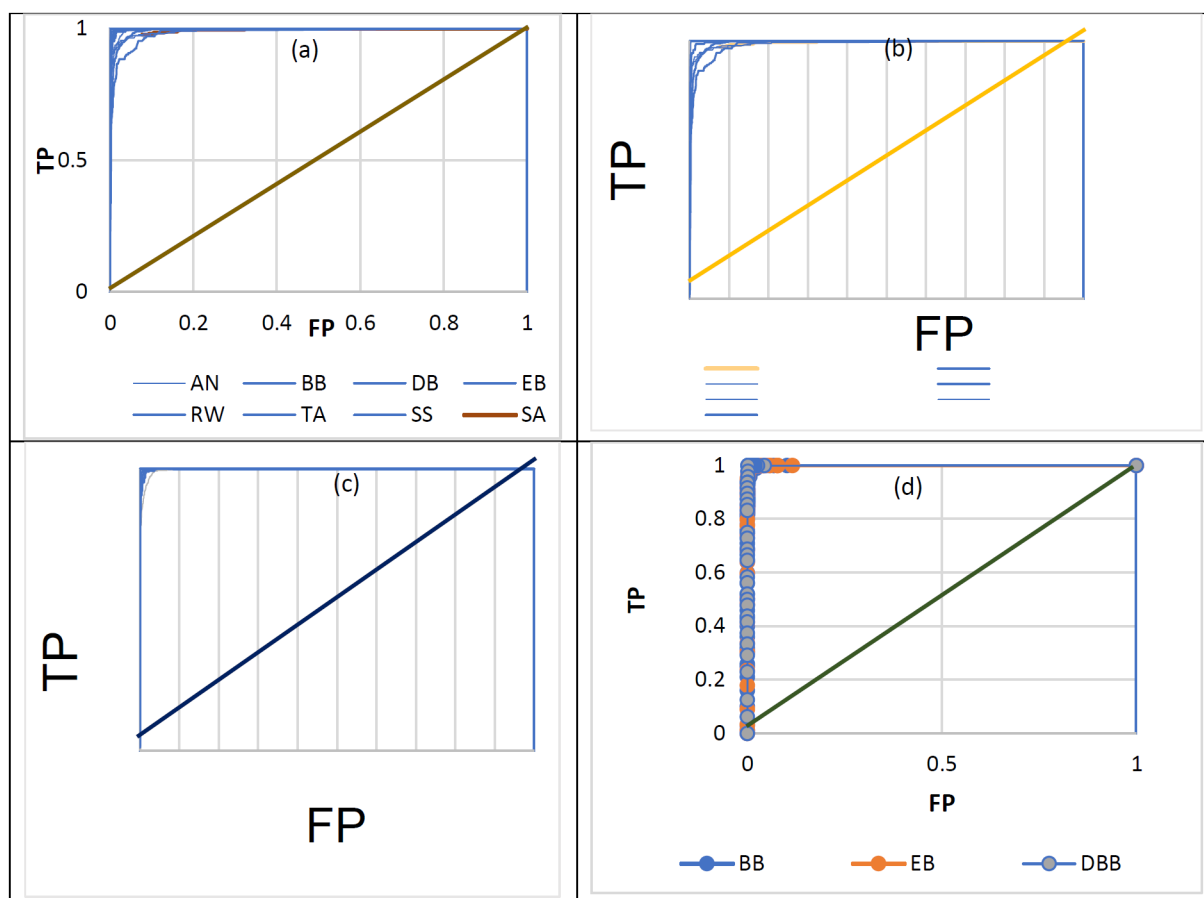


Figure 11—The ROC curves for the RF model concerning flow patterns: a) all patterns, b, and c seven patterns; d) three patterns

Conclusion

This research investigates the effectiveness of ensemble machine learning models, particularly algorithms (LogitBoost, AdaBoost, MultiBoost) and Random Forest algorithms, in classifying two-phase flow patterns within horizontal pipes. The results demonstrate that ensemble models achieve superior prediction accuracy compared to individual classifiers, underscoring the benefits of combining multiple models to enhance accuracy and robustness in flow pattern classification. Volumetric liquid velocity and gas velocity were identified as the most significant features influencing model performance, highlighting their critical role in accurate predictions. The developed models offer efficient and economical solutions for predicting flow regimes in horizontal pipelines, with significant potential for industrial applications, including improved system design, operational efficiency, and safety.

Based on the evaluation measures, it can be concluded that the RF model demonstrated strong performance, with high precision, recall, and F-measure scores, indicating its reliability for classification tasks. However, a detailed comparison of boosting algorithms revealed LogitBoost as the superior performer, achieving the highest accuracy and lowest error rates across multiple metrics. The analysis revealed Logitboost to be the superior performer among the compared algorithms. It consistently achieved the highest accuracy and exhibited the lowest error rates across a range of evaluation metrics. While Multiboost demonstrated competitive performance, following Logitboost closely, Adaboost showed a slight decrease in both accuracy and error rates. The Kappa statistic farther provided additional confirmation

of strong agreement between Logitboost and Multiboot predictions and observed values. These findings suggest that while RF is a robust choice for general classification, LogitBoost outperforms other models in this specific context, making it the preferred option for predictive modeling of horizontal two-phase flow patterns. The results emphasize the importance of algorithm selection based on dataset characteristics and performance metrics, providing actionable insights for industrial applications.

Abbreviations

VI	Superficial velocity for liquid (m/s)
Vg	Superficial velocity for gas (m/s)
AN	Annular flow regime
ST	Surface tension
Den	Fluid Density
Visc	Fluid Viscosity
C	Surfactant concentration
AUC	Area Under the Curve
AN	Annular flow regime
BB	Bubble flow
DB	Dispersed bubble flow
EB	Elongated bubble flow
IW	Inertial wave
PS	Plug slug
PL	Plug flow
PSS	Pseudo slug
R	Ripple
RW	Ripple wave
ST	Surface tension (N/m)
SL	Slug
SA	Slug annular
SS	Smooth stratified
TA	Thin annular
E	Classification accuracy
KS	Kappa statistic
MAE	Mean absolute error
RMSE	Root mean squared error
RAE	Relative absolute error
RRSE	Root relative squared error
RF	Random forest
AdB1	Adaboostm1 algorithm
NBT	Naive Bayes Tree
LWL	Locally Weighted Learning model
LMT	Logical Model Tree
Fm	F-measure or F1-score
MCC	Matthew's correlation coefficient
PRC	Precision Recall Curve
ROC	Receiver Operating Characteristic curve

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